

Multimodal Fusion for Remote Sensing Land-Cover Segmentation

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Abstract—Remote sensing land-cover segmentation is challenging because aerial images often contain visually similar classes and imbalanced land-cover distributions. This project investigates whether image-level textual descriptions can improve segmentation performance compared with a vision-only baseline. A UNetFormer-based segmentation model is extended with a frozen text encoder whose caption embeddings are fused into the segmentation decoder. The experiments compare vision-only and multimodal configurations, different caption variants, CLIP and RemoteCLIP text encoders, auxiliary KL regularization, and composite loss functions including CE+Dice, CE+Lovász, and CE+Tversky. The vision-only UNetFormer baseline achieves 0.6488 test mIoU. Adding textual information through the CLIP-based multimodal model increases test mIoU to 0.6970. The best final configuration, which combines RemoteCLIP with CE+Dice loss, further improves test mIoU to 0.7347, a relative gain of 13.25% over the vision-only baseline. Per-class results and qualitative examples show that the multimodal model improves several difficult and underrepresented land-cover classes such as Built-up and Shrub.

Keywords—*Semantic segmentation, multimodal fusion, vision-language models, remote sensing, class imbalance, region-based loss*

I. INTRODUCTION

Remote sensing images are widely used for land-cover mapping, urban analysis, agricultural monitoring, and environmental observation. Semantic segmentation is a fundamental task in remote sensing because it provides pixel-level understanding of the Earth's surface, enabling accurate identification and mapping of land-cover classes for environmental monitoring and resource management.

Despite its importance, remote sensing land-cover segmentation remains challenging due to the visual complexity of aerial images. Different land-cover classes may have similar textures, colors, and spatial patterns. Vision-only segmentation models can learn strong visual representations, but they may still struggle when local visual evidence is ambiguous or when some classes are underrepresented. Image-level textual descriptions can provide additional semantic context, which makes multimodal segmentation a promising direction for improving land-cover prediction.

Recent studies have shown that textual information and vision-language representations can improve remote sensing segmentation. However, existing work mainly focuses on multimodal alignment or adapting pretrained representations to the remote sensing domain. There is still a need to evaluate how different textual description variants, text encoders, and training objectives affect segmentation performance under the same experimental framework.

The objective of this project is to design and evaluate a multimodal framework for remote sensing land-cover segmentation. The main research question is whether

incorporating image-level textual descriptions improves segmentation performance compared with a vision-only baseline. The task is defined as semantic segmentation, where each pixel is assigned to one of seven land-cover classes. A UNetFormer-based model is used as the main visual architecture and extended with a text encoder whose caption embeddings are fused into the segmentation decoder. The main experiment compares the vision-only baseline with the multimodal model to answer this question. Ablation studies then examine which design choices drive the benefit, namely the caption source, an auxiliary regularizer, the text encoder, and the loss function. Model performance is assessed using mean Intersection over Union (mIoU), per-class Intersection over Union (IoU), and overall accuracy (OA).

II. LITERATURE SURVEY

Recent studies in remote sensing segmentation show that vision-only models can struggle with complex scenes, intra-class variation, and limited semantic understanding [1]. Therefore, several studies have explored multimodal segmentation methods that use textual information to support visual feature learning. TAMS and FSVLM represent methods that generate textual descriptions and integrate them into segmentation. TAMS adopts a UNetFormer-based architecture with a BERT text encoder and aligns image and text features using cross-attention and contrastive learning [2]. FSVLM, applied to farmland segmentation, combines a segmentation model with a vision-language framework. It shows that textual descriptions can capture spatial, environmental, and temporal relationships that are difficult to model using visual features alone. It also reports that the level and content of textual descriptions affect segmentation performance [3].

CLIP2RS and RemoteCLIP follow another direction by using pretrained vision-language representations. CLIP2RS adapts CLIP for remote sensing segmentation with textual prompts. It also uses a dual-granularity alignment strategy that aligns both image-level and pixel-level features. This strategy helps address intra-class variation and class imbalance [1]. RemoteCLIP takes a related approach, unifying heterogeneous annotations into image-text pairs. It then uses these pairs to train a domain-adapted vision-language foundation model [4].

Overall, these studies show that textual information and vision-language representations can support remote sensing segmentation. However, existing studies mainly focus on proposing multimodal alignment strategies or adapting pretrained representations to the remote sensing domain. Less attention is given to controlled comparisons of different caption variants, text encoders, and training objectives within the same segmentation framework. Therefore, a controlled experimental setting is needed to isolate the effects of caption content, text encoder choice, and training objective on segmentation performance.

III. METHODS

A. Dataset

The dataset contains 10,000 aerial RGB images at 256×256 resolution. Each image has a pixel-level mask over seven land-cover classes, namely Tree, Shrub, Grass, Crop, Built-up, Barren, and Water. Each image is also paired with five caption variants drawn from three source types, text-only, hybrid, and vision-only, generated by different language and vision-language models. The class distribution is strongly imbalanced, with Grass covering about 45% of pixels on average and Shrub only about 0.8%. RGB mask colors are converted to class-index masks before training, following the dataset's defined color-to-class mapping. The data is split into 7,000 training, 1,500 validation, and 1,500 test images, stratified by the dominant class of each image, and the same split is used in every experiment.

B. Model Architecture

The visual backbone is selected by comparing two vision-only baselines from different architectural families. UNetFormer uses a ResNet-18 CNN encoder with a global-local transformer block decoder, a feature refinement head, and an auxiliary head for deep supervision, at about 12 million parameters. Swin-UperNet uses a Swin-Tiny transformer encoder with a UPerNet decoder, at about 60 million parameters. UNetFormer is selected as the backbone for the multimodal model, as it gives the stronger baseline despite its smaller size.

The multimodal model extends UNetFormer with a text branch. A frozen pretrained text encoder, CLIP with a ViT-B-32 architecture, maps each caption to a 512-dimensional L2-normalized embedding. No gradients flow into the encoder. The embedding enters the decoder through gated cross-attention at each decoder stage. The gate is initialized at zero, so text has no influence at the start of training and the model behaves like the vision-only baseline. The gate is learned, so the model controls how much text to admit. This adds text gradually without disturbing the pretrained visual features.

C. Training and Evaluation

All models share one recipe, namely 30 epochs, AdamW, learning rate $6e-4$, weight decay 0.01, cosine annealing schedule, and batch size 8. The base objective is weighted cross-entropy, with an auxiliary head loss added for deep supervision. Class weights follow median frequency balancing, where each weight is the median class frequency divided by that class's frequency, raising the contribution of rare classes. Each configuration is checkpointed at its best validation mIoU and selected on validation, then evaluated once on the test set. The primary metric is mean Intersection over Union, mIoU, with per-class IoU and overall accuracy as secondary metrics.

D. Experimental Design and Ablations

The main experiment compares the multimodal model against the vision-only UNetFormer baseline under the same training setup, so that the difference reflects the addition of text. This answers the main research question of whether textual descriptions improve segmentation.

A set of ablation studies then examines which design choices drive the benefit. Each study varies one component, namely the caption source, a KL-divergence auxiliary regularizer, the text encoder, or the loss function, and holds the others fixed, so that each change can be attributed to that component.

1) Caption Source

The multimodal model is trained on each of the five caption variants and compared against the vision-only baseline. This tests which caption type, text-only, hybrid, or vision-only, contributes most to segmentation performance. A separate check addresses whether the model relies on numbers stated in some captions, such as a grass percentage, rather than on the described scene. The text-only and hybrid captions, which contain percentages, are retrained with the numbers removed and the descriptions kept. A small change would show that the benefit comes from semantic content, not numerical proportions.

2) KL-Divergence Auxiliary Regularizer

A Kullback-Leibler term is added to the model with the best caption from the caption source analysis. The term aligns the predicted class distribution over all pixels with the ground-truth distribution of the image. It is weighted at 0.5 and applied only during training, not at inference. This tests whether explicit supervision of global class proportions helps beyond the textual information.

3) Text Encoder

The CLIP encoder is replaced with RemoteCLIP, a variant fine-tuned on remote sensing image and text data, and evaluated across all five captions. Both share the ViT-B-32 architecture, so only the encoder weights change. This tests whether a domain-adapted encoder yields more useful caption embeddings, and whether the best caption is stable across encoders.

4) Loss Function

Weighted cross-entropy is replaced with a region-based composite loss, evaluated as three variants on the best caption from the caption source analysis. Dice optimizes region overlap, Lovász-Softmax is a smooth surrogate for the Jaccard index that is the metric itself, and Tversky generalizes Dice with asymmetric penalties biased toward recall, which benefits rare classes. Each region loss is added to weighted cross-entropy with both terms weighted at 0.5. Cross-entropy provides a stable per-pixel gradient while the region term directly optimizes overlap.

The final model combines the best-performing encoder (RemoteCLIP) and loss (CE+Dice) from the ablations on the best caption (text_qwen3-4b).

IV. RESULTS

A. Baseline Selection

The two vision-only baselines reach a validation mIoU of 0.6580 for UNetFormer and 0.6396 for Swin-UperNet. UNetFormer performs better on all seven classes and is selected as the vision-only baseline. On the held-out test set it reaches 0.6488, which serves as the reference for the multimodal comparison.

B. Effect of Textual Information

Adding textual information improves segmentation over the vision-only baseline. The best caption, text_qwen3-4b, raises test mIoU from 0.6488 to 0.6970, a relative gain of 7.42%, and overall accuracy from 0.8621 to 0.8894. This answers the main research question. The improvement is not uniform across captions, however. Most variants improve over the baseline, while the vision-only captions provide negligible benefit, one even falling slightly below the baseline, which motivates the caption analysis that follows.

C. Ablations

1) Caption Source

The five caption variants fall into two groups by how they are produced, and this split maps directly onto their effect. The text-only and hybrid captions, which draw on structured class information, all improve over the baseline. The text-only caption `text_qwen3-4b` performs best at 0.6970, a relative gain of 7.42%, and the two hybrid captions, `hybrid_gemma3-4b` and `hybrid_qwen3-vl-8b`, follow closely at 0.6930 and 0.6906, gaining 6.81% and 6.44%. The two vision-only captions, `vision_gemma3-4b` and `vision_qwen3-vl-8b`, are produced from the image alone and behave differently, giving little or no gain at -1.46% and +0.48%. The improvement therefore depends on the caption type rather than on the presence of text. The percentage-removal check refines this further. Removing the numerical percentages from `text_qwen3-4b` and the two hybrid captions changes test mIoU by at most 0.81%, so the benefit comes from the qualitative semantic content of the captions rather than from numerical class proportions.

2) KL-Divergence Auxiliary Regularizer

The Kullback-Leibler regularizer gives a small gain. Added to `text_qwen3-4b`, it raises test mIoU from 0.6970 to 0.7046, a relative gain of 1.09% over the multimodal baseline. This gain is later exceeded by the loss ablation, so the regularizer is not used in the final model.

3) Text Encoder

Replacing CLIP with RemoteCLIP gives a small and consistent gain across captions. RemoteCLIP improves four of the five variants and regresses only on `vision_qwen3-vl-8b`, with all changes between -0.0040 and +0.0057 mIoU. On the best caption, `text_qwen3-4b`, it raises test mIoU from 0.6970 to 0.7008, a relative gain of 0.55% over the multimodal baseline. The caption ranking is unchanged, with `text_qwen3-4b` remaining strongest under both encoders. The gain on `text_qwen3-4b` concentrates almost entirely on the Built-up class, which improves by 0.0238 IoU.

4) Loss Function

The region-based loss gives the largest single-component gain. All three composite losses are evaluated on `text_qwen3-4b` and improve over weighted cross-entropy, performing similarly to each other. CE+Dice reaches 0.7281, CE+Tversky 0.7279, and CE+Lovász 0.7275, with relative gains of 4.46%, 4.44%, and 4.38% over the multimodal baseline. CE+Dice is selected on validation. The improvement concentrates on the hardest classes, with Built-up rising by 0.1223 IoU and Shrub by 0.0455 IoU.

D. Final Model

The final model combines the two strongest ablation components, the RemoteCLIP encoder and the CE+Dice loss, on `text_qwen3-4b`. It reaches a test mIoU of 0.7347, a relative gain of 5.42% over the multimodal baseline and 13.25% over the vision-only baseline. The two components contribute almost independently. Their combined gain exceeds the sum of the two individual gains by only 0.0028 mIoU, so applying them together is approximately additive.

E. Overall Comparison

Table I summarizes the test mIoU at each stage from the vision-only baseline to the final model. The test mIoU increases at every stage. The loss change gives the largest single-component gain, and the final model, which combines the RemoteCLIP encoder with the CE+Dice loss, reaches the highest value at 0.7347.

TABLE I. TEST mIoU ACROSS CONFIGURATIONS

Configuration	mIoU	Δ mIoU (%)
Vision-only baseline, UNetFormer	0.6488	-
Multimodal, CLIP	0.6970	+7.42%
Multimodal, CLIP, with KL regularizer	0.7046	+8.60%
Multimodal, RemoteCLIP	0.7008	+8.01%
Multimodal, CLIP, with CE+Dice loss	0.7281	+12.22%
Final, RemoteCLIP, with CE+Dice loss	0.7347	+13.25%

Table II reports per-class IoU for the vision-only baseline, the multimodal CLIP model, and the final RemoteCLIP + CE+Dice model.

TABLE II. PER-CLASS IOU ON THE TEST SET

Class	Vision-only	Multimodal	Final
Tree	0.8608	0.8758	0.8830
Shrub	0.2476	0.3261	0.3708
Grass	0.7631	0.8057	0.8279
Crop	0.7416	0.8159	0.8228
Built-up	0.5631	0.5893	0.7137
Barren	0.4869	0.5665	0.6068
Water	0.8784	0.8994	0.9182
mIoU	0.6488	0.6970	0.7347

The per-class values reveal where each stage acts. Adding text raises every class, with the underrepresented classes benefiting most. The encoder and loss changes then act most strongly on Built-up, which rises from 0.5893 to 0.7137, while the already strong classes Tree, Grass, and Water change little throughout. Shrub remains the weakest class at every stage, reaching only 0.3708 in the final model.

F. Qualitative Comparison

Fig. 1 follows one test scene from input to prediction. The same input image is used for all three models, and the same caption is supplied to the two multimodal models, so the panels show how the output mask changes as textual information and the region-based loss are added.

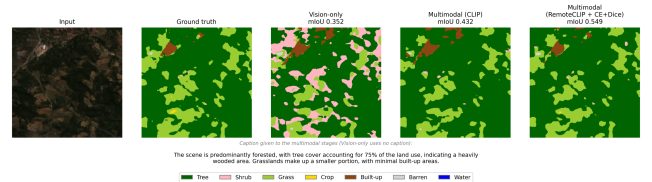


Fig. 1. Qualitative comparison on a representative test scene

The first two panels show the input image and the ground-truth mask. The next three show the predicted masks from the vision-only baseline, the multimodal CLIP model, and the final model with RemoteCLIP and CE+Dice. The text below the panels is the caption given to the multimodal models. Reading left to right, the vision-only output scatters Shrub across the forest. Adding the caption removes most of the spurious Shrub and recovers much of the Grass, while the final model improves the Built-up patch and recovers the Crop region, sharpening the result. The per-image mIoU rises from 0.352 to 0.432 to 0.549 across the three models.

V. DISCUSSION

Textual descriptions improve segmentation when they carry information the image alone does not provide. For a scene that is 92% grass, the text-only caption describes it as predominantly grass, while the vision-only caption for the same scene reports a rugged, arid landscape of rocky terrain and exposed bedrock. The vision-only caption misreads the aerial view and names the wrong land cover, so it gives the model no reliable information about the classes present. This is systematic. Vision-only captions fail to mention Grass in over 94% of the images where it is substantially present, while text-only and hybrid captions rarely omit it. The percentage-removal check agrees, changing the result by at most 0.81%, so the benefit comes from the meaning of the caption rather than from stated proportions.

This pattern is consistent with prior text-assisted segmentation work. TAMS adds textual descriptions to a UNetFormer backbone on a class-imbalanced dataset and reports that text improves segmentation [2]. FSVLM studies the content of the descriptions and finds that captions describing the relevant attributes help more than less informative ones [3]. Both studies and the present result point to the same conclusion. What matters is whether the text carries reliable information about the scene, not whether text is present at all.

The improvements help most where the vision-only model is weakest. Built-up, Shrub, and Barren, the three lowest-scoring classes at the baseline, improve substantially, while the strongest classes, Tree and Water, change little. Shrub gains the most in relative terms but stays the lowest-scoring class in the final model, so the rarest class is helped without being solved. The two refinements that form the final model act on different weaknesses. The region-based loss gives the single largest gain, and it acts mostly on Built-up. This makes sense, because the region-based loss directly optimizes overlap, so it helps most on the small classes that cross-entropy handles poorly. The RemoteCLIP encoder, by contrast, adds little on its own, which matches RemoteCLIP's report of small and uneven gains over CLIP across tasks [4]. Because the encoder and the loss fix different weaknesses, their gains do not overlap, and applying both together gives almost exactly the sum of applying each alone, with a difference of only 0.0028 mIoU.

The main limitation comes from the model's reliance on the dominant class named in the caption. When a caption states that one class covers most of the scene, the model can over-predict that class and erase the smaller classes that are actually present. Fig. 2 shows this on a dense urban scene whose caption states an 88% built-up fraction. The multimodal CLIP model fills almost the entire scene with Built-up and loses the smaller Grass and Tree regions, dropping below the vision-only baseline on this scene. The same reliance on caption content that drives the overall gains is therefore also the source of the main failure. This kind of error in complex scenes is reported by prior work as well, with FSVLM noting that segmentation errors arise in complex scenes [3] and TAMS reporting that its hardest classes stay difficult even with text [2].

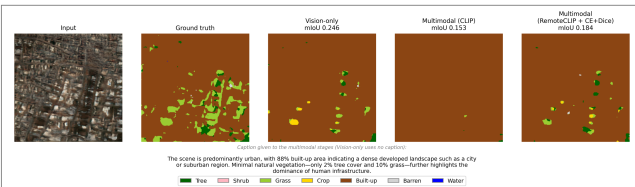


Fig. 2. A failure case

Several directions could address this limitation. The failure is not that the caption omits the minority classes. In Fig. 2 the caption names both tree and grass cover, yet the model still erases them. The real issue is different. A caption describing one class as overwhelmingly dominant pulls the prediction toward that class. The gated cross-attention could be constrained to fix this. It would let text adjust the visual prediction rather than replace it. This keeps the visual evidence for small classes from being overridden. Calibrating the gate to weight text against visual confidence is a natural next step for complex scenes.

In summary, textual descriptions improve land-cover segmentation when they carry reliable information about the scene, and the largest single gain comes from a region-based loss that targets the underrepresented classes directly. The encoder and loss changes act on different weaknesses, so their benefits combine almost additively. The same dependence on caption content that produces these gains is also the main limitation, since a caption dominated by one class can suppress the smaller classes that are present.

VI. CONCLUSION

This work studied whether textual descriptions improve land-cover segmentation under severe class imbalance. A UNetFormer backbone was extended with a frozen vision-language text encoder and gated cross-attention. The design was examined through ablations on the caption source, an auxiliary regularizer, the text encoder, and the loss function. Reliable textual descriptions improved segmentation over a vision-only baseline, and the caption-source ablation showed that the benefit depends on the semantic content of the caption rather than on the presence of text. Among the design choices examined, the region-based loss was the change that most affected the result, giving the largest single gain. The final model, combining a domain-adapted encoder with this loss, reached a test mIoU of 0.7347, a relative improvement of 13.25% over the vision-only baseline. The encoder and the loss act on different weaknesses, so their gains combine almost additively rather than overlapping.

The gains concentrated on the underrepresented classes, where visual features alone are least reliable, which is where added textual information and direct optimization of region overlap are most valuable. The main limitation is that a caption dominated by one class can suppress the smaller classes that are present, so the same dependence on caption content that drives the gains also produces the principal failure. Addressing this through a tighter balance between textual and visual guidance is the main direction for future work.

VII. REFERENCES

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